



Procedure for Installation, Operation and Maintenance Manual SDP-6 / SDP-8



Ternetangen 65
N-5420 Rubbestadneset

00007464, rev. K

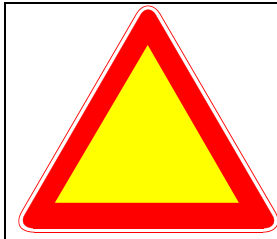
Revision List

Rev	Date	Description	Author	Reviewed	Approved
X1	11.12.01	REVIEW	ESE		
0	21.12.05	AFC	HKA	KH	ESE
A	17.01.08	New logo and layout, inserted table of content	KAA		
B	10.12.2009	Revised, to cover digital SDP-6 sensors	SM	SM	Tkg
C	27.01.2010	Revised, to cover Level Monitoring	SM	SM	TKG
D	11.08.2011	Revised, Item 5	SM	RM	TKG
E	06.02.2013	Changed Logo	CS	MN	RM
F	25.06.2015	Changed layout to TC document, updated text.	BH	CF	AE
G	19.11.2015	Improved content	BH	CF	AE
H	24.08.2017	Changed section 8	BH	TKG	EV
I	22.06.2020	Revised to include SDP-8	EOH	TKG	BH
J	02.09.2021	New logo	EOH	BHV	CF
K	15.03.2022	Item 8.2 Storage temperature changed to -40 to +70 °C.	EOH	EN	AE

TABLE OF CONTENTS

1	GENERAL	4
2	PURPOSE	4
2.1	GENERAL	4
2.2	PACKING AND TRANSPORTATION	4
3	HEALTH AND SAFETY STATEMENTS.....	5
4	TESTING	6
4.1	FUNCTIONAL TESTING.....	6
4.2	EQUIPMENT	6
4.3	HOOK UP.....	7
4.4	VERIFICATION OF SENSOR READINGS.....	7
5	INSTALLATION.....	8
5.1	INSTALLATION POSITION AND DIRECTION	8
5.2	INSTALLATION OF REMOTE SEALS	8
5.3	ROUTING OF THE CAPILLARY TUBES	8
5.4	SDP USED FOR FLOW MONITORING	8
5.5	SDP USED FOR LEVEL MONITORING	9
5.6	INTRODUCTION	9
5.7	REQUIRED INFORMATION	9
5.8	ZERO ADJUSTMENT	9
5.9	CAPILLARY TUBES.....	10
5.10	MEMBRANE SYSTEM.....	10
5.11	FIXING TOOLS.....	10
5.12	WELDING.....	10
5.13	PAINTING ON STRUCTURE	11
6	OPERATION INSTRUCTION	11
7	MAINTENANCE.....	11
8	STORAGE	12
8.1	GENERAL CONDITIONS	12
8.2	STORAGE.....	12

1 GENERAL



WARNING: Ensure that this Installation, Operation and Maintenance manual is appropriate for the instrument / sensor(s).

2 PURPOSE

2.1 GENERAL

The purpose of this Manual is to ensure that the Installation, Operation and Maintenance are carried out properly in order to ensure the customer of the Subsea sensor's conformance to the specifications given in The Procurement Specification.

This is a generic procedure covering several different types and combinations of Siemens Energy AS, Subsea differential pressure sensors hereafter called SDP.

2.2 PACKING AND TRANSPORTATION

Avoid excessive shock and vibrations during transport and handling, even though the equipment itself is resistant towards shock and vibration, this is only true when the units are properly fixed and secured.

The SDP sensor consists of 1 sensor body and 2 flanges connected by a capillary tube. A Special transport case is prepared for packing, storing and transport of the SDP. Inside the box the sensor 3 parts are bolted onto a metal plate and we strongly advise that the units are kept secured in this box/metal plate until the final installation destination is reached.

The transportation case must always be in its upright position (see marking on the case).

The SDP considerable high own weight makes a dangerous combination if the transportation case is handled roughly (dropped, turned from upright positions etc).

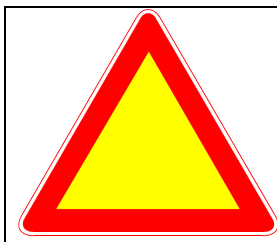
When the unit is to be lifted from the transportation case, use a lifting crane and the supplied hooks, never attempt to lift the unit by hand.

Due to the weight difference between the vulnerable capillary tubes, remote seals and the main body of the sensor, the capillary tubes are at high risk for damage during installation and handling, special care is therefore required.

Be sure that the SDP is stored in clean environment and preferably at room temperature. If the sensor is stored more than six - 6 - months before installation, a test of the sensor signal at ambient pressure and temperature should be carried out.

If the results from the testing show any unexpected readout from the sensor contact Siemens Energy for assistance.

2.2.1 SAFETY



WARNING: Only personnel deemed qualified shall operate equipment involving high pressures and temperature. Protection glasses, gloves and shoes are required.

Other PPE to be used as appropriate for the area and work to be performed, see company safety instructions.

3 HEALTH AND SAFETY STATEMENTS

It is recommended that the Client performs a "Risk Assessment" for all aspects of the Sensor Installation from initial delivery to final dispatch, to identify the hazards associated with the work.

Because of the considerable high weight of each Sensor Unit, those staff preparing to move a Sensor Unit from the Delivery Box must ensure that they "Assess the Identified Hazards" for each movement of the Sensor.

The Sensors do not contain any chemicals or compositions that could be hazardous upon exposure. Electrical jumpers can contain silicon oil or similar, although this oil is not hazardous to health, any spillage should be removed immediately to avoid slip hazards.

During Soldering Operations, it is recommended that fumes are removed from the immediate working area to avoid inhalation by the operator.

Care in the selection of the correct tools for the final "Torque Setting of the Bolts" is important to avoid accidents through damaged/poorly maintained tooling.

Manual Handling, Lifting and Carrying are known to be the largest contributors to accidents. Ensure that mechanical handling aids are used whenever possible to avoid manual handling. Where manual handling is considered appropriate for the task safe lifting guidelines must be followed, e.g. adopt correct posture, consider team lifting,

employ safe lifting technique, etc. Ensure positions of lifting eyes keeps assembly in balance.

Figure 1 shows the correct slinging method for an SDP-6 with Remote Seal Flanges.

Figure 1.



4 TESTING

4.1 FUNCTIONAL TESTING

During site receive check, perform a Functional Test to ensure the SDP is working properly.

4.2 EQUIPMENT

For analog 4–20 mA output, a loop calibrator can be used.

For HART Protocol, a standard handheld HART Communicator can be used.

For other digital outputs such as Modbus or CANopen, a control system or a laptop with software package and a converter is necessary to be able to read the sensor.

There are numerous commercial software and converter units available on the market.

Examples of software tools:

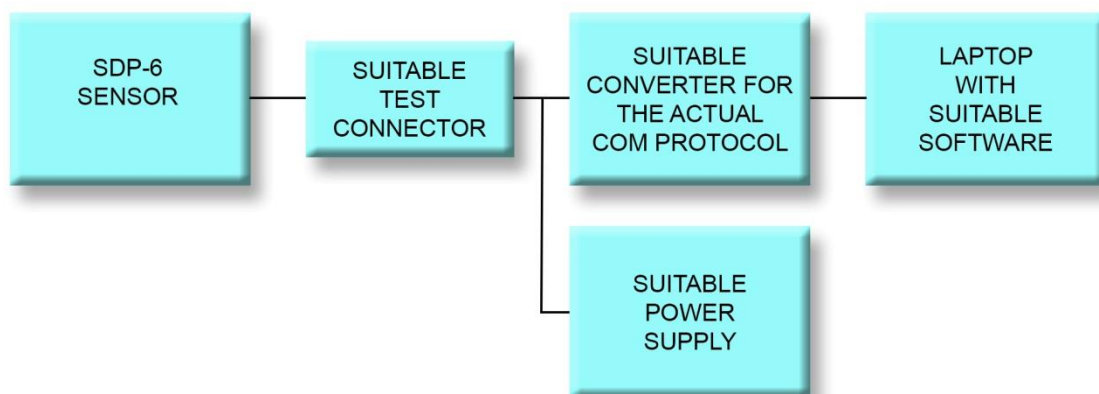
- CANopen: CANopenTester.exe
- RS485 Modbus: Win-Tech Modscan32.exe, <http://www.win-tech.com/html/modscan32.htm>
- RS485 KOS: KosViewer.exe

Examples of converters:

- CANopen: Ixxat converter for high speed or fault tolerant CAN:
<http://www.ixxat.com>
- Rs485: MOXA interface converter RS232-Rs485
<http://www.moxa.com/product/tcc-100.htm>

A suitable test connector, a test cable and power supply are also needed.
Figure 3 shows the different elements needed to hook up a testloop with SDP-6.
SDP-8 hook up is identical.

Figure 3.



4.3 HOOK UP

Hook up the sensor according to the information given in the GA drawing.

4.4 VERIFICATION OF SENSOR READINGS

Verify that the readings from the sensor are in accordance with the criteria given in Specification sheet for the actual project.

Note:

The SDP is calibrated and tested at a calibration laboratory in a controlled environment and the reference instruments used are of high accuracy. The SDP could also have been subject to High Static Calibration and therefore calibrated with specific pressure and temperature applied.

It can be difficult or even impossible to recreate these conditions (e.g. pressure and temperature) during testing at site and therefore this needs to be taken into consideration during evaluation of the sensor readings.

The acceptance criteria for measurements taken onsite in ambient conditions needs to reflect the expected offset of the SDP in ambient condition.

If more clarification is needed contact Siemens Energy for further information.

5 INSTALLATION

5.1 INSTALLATION POSITION AND DIRECTION

The main SDP body can be installed in any direction or any position with respect to the Remote Seals.

The SDP pressure inlets, remote seal, are marked with "L" and "H" for Low pressure side and High-pressure side respectively. Be sure to route capillary tubes and install remote seal to their correct pressure side.

5.2 INSTALLATION OF REMOTE SEALS

The Remote Seals will either be bolted or welded to the meter body or the tank, ref GA drawing for the project. If welded Remote Seals, a welding procedure will be stated for the project and if bolted Remote Seals the assembly must be according to the flange standard defined by the actual project.

5.3 ROUTING OF THE CAPILLARY TUBES

For the purpose of handling and transport, the capillary tubes are coiled. During installation this coil must be straightening out for the remote seals to reach their installation position.

Straightening out the coils must be done with care to avoid a sharp bend.

The capillary tubes must be routed and fixed such that the environmental influences become equal for the entire length of the tubes (both high pressure- and low-pressure side). Failing to do so will degrade the total accuracy of the sensor.

It is advised that the capillary tubes are supported and fixed every 200mm. Make sure that the supporting bracket does not introduce temperature to the capillary tube from other parts of the construction. This may have a detrimental effect to the sensor readout accuracy.

5.4 SDP USED FOR FLOW MONITORING

When the SDP is used for Flow monitoring, the sensor will be assembled to a Flow meter, e.g. a Venturi or a V-Cone. The H- port of the SDP will be assembled to the High-Pressure port and the L- port will be assembled to the Low-Pressure port of the Flowmeter. The GA drawing approved for the actual project will show the details due of the assembly.

5.5 SDP USED FOR LEVEL MONITORING

5.6 INTRODUCTION

The SDP can be used for Level Monitoring of tanks, etc. The high-pressure port must be connected to the low-level position of the tank and the low-pressure port must be connected to the high-level position of the tank. The weight of the actual fluid in the tank will generate a differential pressure between high and low position of the tank. The SDP will measure this difference pressure which can be used as a measurement for the fluid level in the tank.

5.7 REQUIRED INFORMATION

The height difference between high- and low-pressure port must be specified, see illustration 1 below. This information is to be found in the Procurement specification.

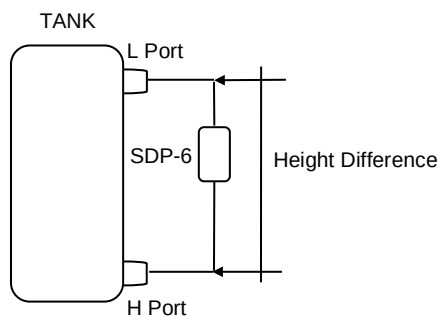


ILLUSTRATION 1

5.8 ZERO ADJUSTMENT

For projects where the SDP will be used for Level Monitoring, Vertical Flowmeter measuring gas or others where H&L ports are at different height a Zero adjustment needs to be considered.

For Flowmeter and other applications where the height difference is specified the adjustment can normally be completed as a part of the FAT.

When the SDP is used for Level Monitoring on e.g. vessels where the high difference is significantly the best result is achieved by performing the zero adjustment “on site” after the sensor is installed.

Note:

Functional Test of SDP which is Zero adjusted according to item 5.8, with the high- and low pressure port placed at the same level will have an offset reading representing the specified height difference.

5.9 CAPILLARY TUBES

The Capillary tubes must be handled with extreme care, do not:

- Lift or carry the Remote Seal or the Housing by the capillary tubes
- Bend the capillary tubes more than absolutely necessary / required for facilitating the installation.
- Make any bend on the capillary tubes with radius less than 30mm and always use appropriate tool when making the eventual bends
- Do not climb on, step on or use the capillary tube as a climbing aid.

Failing to do so may permanently damage the complete sensor or degrade the sensors accuracy.

5.10 MEMBRANE SYSTEM

The Membrane system must be handled with extreme care, do not touch or come near the membrane surface by hand or any object.

Note that the membrane is very thin and fragile, even the slightest contact with sharp object may permanently damage the complete sensor.

5.11 FIXING TOOLS

Never use air tool when fixing the Remote Seal and Main body bolts. Avoid other types of shock and vibration impacts.

5.12 WELDING

At any time during any type of welding operations on or in vicinity of the dP-sensor the SDP need to be grounded.

Best practice for grounding the SDP is:

For SDP supplied with no flying leads the sensor is grounded by spring between the pins and the sensor housing. This spring is supplied and mounted by Siemens Energy before shipping.

For SDP with flying leads, twist the end of the cables together and ensure that they are adequately connected to the sensor housing.

For SDP supplied with connector best practice is to use a testconnector with flying leads. Twist all leads of the testconnector and ensure that they are adequately connected to the sensor housing.

5.13 PAINTING ON STRUCTURE

If the sensor is supplied un-painted, the sensor can be prepared and painted along with the host structure. During surface preparation, e.g. blasting, Pin Hole Tests or other test methods that utilising an electrical method the SDP needs to be grounded as described in section 5.12.

Special consideration should be taken towards masking of the electrical connector.

6 OPERATION INSTRUCTION

The operating of the unit is straight forward and easy to use. The output from the SDP is an industry standard of 4- 20 mA signal/HART or digital outputs as MODBUS or CANOpen.

The SDP will work together with the supply/receiver circuitry in the receiving module. If the receiving module is not available the SDP can communicate via PC software and interface hardware, see section 4.2. For more information contact Siemens Energy. For 4-20 mA, a standard loop tester can be used. For HART Communicator, a standard handheld communicator can be used.

After installation subsea, the signal from the SDP will form a part of the client's main system, and all operational instructions should be according to the client's procedures.

7 MAINTENANCE

Generally, no maintenance is required on stored assemblies. However, assemblies that have been in storage should be inspected prior to installation.

After installation subsea, it will normally not be possible to carry out any maintenance of the dP-transmitter.

If the SDP for any reason should be retrieved after installation, the sensor should be controlled and/or considered returned to the manufacturer for a complete check/recalibration.

8 STORAGE

8.1 GENERAL CONDITIONS

Be sure that the SDP is stored in clean and air-dry environment and preferably at room temperature.

Humidity:	Normal - up to 100%.
Chemicals:	Avoid contact with mineral oils, Corrosive gas or acid.
Mechanical:	Avoid impacts, footsteps or sharp edges.
Others:	Avoid storage near electrical motors, batteries, or welding equipment. Necessary measures should be taken to avoid that dust, or any other foreign particles, accumulate on the assemblies. Assemblies should also be protected against rodents if necessary.

8.2 STORAGE

It is recommended during storage that the assemblies be stored in the original packing crate. Connectors, sensors and transmitters must be protected with bubble plastic or equivalent wrapping. If there is sunlight present in the storage area, cover the assemblies with black plastic or similar protection.

- Recommended temperature: -40 to +70°C
- Light: No direct sunlight, preferably dark